

Thermodynamics I: Sample Exam Questions

I: True or false questions

Ex. 1. One corollary of the second law of thermodynamics states that the change in entropy of a closed system must be greater than or equal to zero.

Ex. 2. Intensive properties may be functions of both position and time, whereas extensive properties can vary only with time.

Ex. 3. The entropy of a fixed amount of an incompressible substance increases in every process for which temperature increases.

Ex. 4. A spring is compressed adiabatically and its internal energy increases.

Ex. 5. The Clausius statement of the second law denies the possibility of transferring energy by heat from a cooler to a hotter body.

Ex. 6. For a gas modeled as an ideal gas, the specific internal energy, enthalpy, and entropy all depend on temperature only.

Ex. 7. The change in specific volume from saturated liquid to saturated vapor, $(v_g - v_f)$, at a specified saturation pressure increases as the pressure decreases.

Ex. 8. Kinetic and potential energy each are extensive properties of a system.

II: Multiple-choice questions

Ex. 1. Regarding the equation $q = \int_1^2 c_v dT + \int_1^2 p dv$, which of the following statement is the MOST appropriate.

- a. *It applies to reversible processes of ideal gases in a closed system.*
- b. *It applies to reversible processes of real gases in an open system.*

- c. It applies to any processes of any gas in a closed system.
- d. It applies to any processes of any gas in an open system.

Ex. 2. Which of the following can be considered an ideal gas?

- a. high-pressure gas inside a diesel engine
- b. water steam inside a gas turbine
- c. water vapor contained in air inside an air-conditioning system
- d. refrigerant gas inside a refrigerator compressor

Ex. 3. For air undergoing *isobaric* ($p=\text{const}$) heating, how much heat can be converted to increases of internal energy?

- a. 37%
- b. 65%
- c. 68.4%
- d. 71.4%

Ex. 4. If the entropy of a substance is proportional to its absolute temperature, T , then its heat capacity must be proportional to {blank}

- (A) $1/T$
- (B) constant (T^0)
- (C) T
- (D) T^2

Ex. 5. The critical point of a real gas is {blank}

- (A) where the gas changes to a liquid.
- (B) where the isotherm curve in a P - v diagram goes horizontal.
- (C) the point where gas, liquid and solid can coexist in equilibrium.
- (D) the pressure and volume on the critical temperature isotherm curve where it goes critical.

III. Computational problems

Ex. 1. The van der Waals equation of state is given by

$$\left(P + \frac{a}{v^2}\right)(v - b) = RT$$

where P is the pressure, v the specific volume, R the gas constant and T the temperature.

(1) Derive expressions for the critical temperature T_{cr} and the critical pressure P_{cr} in terms of R and the two characteristic constants of a and b .

(2) The thermal expansion coefficient is defined as

$$\kappa_T = -\frac{1}{v} \left(\frac{\partial v}{\partial P} \right)_T$$

and the isothermal compressibility as

$$\beta = \frac{1}{v} \left(\frac{\partial v}{\partial T} \right)_P$$

Find expressions for κ_T and β in terms of v , T , R , a , and b .

Hint: $\left(\frac{\partial v}{\partial P} \right)_T \left(\frac{\partial P}{\partial T} \right)_v \left(\frac{\partial T}{\partial v} \right)_P = -1$

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Ex. 2. In order to cool 1 ton of water at 20°C in an insulated tank, a person pours 140 kg of ice at -5°C into the water. Determine (a) the final equilibrium temperature in the tank and (b) the entropy generation during this process. Hint: Take the ice and the water as one isolated system.

Properties The specific heat of water at room temperature is $c = 4.18 \text{ kJ/kg } ^{\circ}\text{C}$, and the specific heat of ice at about 0°C is $c = 2.11 \text{ kJ/kg } ^{\circ}\text{C}$. The melting temperature and the heat of fusion of ice at atmospheric pressure are 0°C and 333.7 kJ/kg , respectively.

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Ex. 3. An insulated 0.15-m^3 tank contains helium at 3 MPa and 403 K. A valve is now opened, allowing some helium to escape. The process can be generally regarded as a *uniform-flow process*, where constant average properties are assumed of the helium leaving the tank. The valve is closed when one-half of the initial mass has escaped. Determine the final temperature and pressure in the tank.

Properties The specific heat ratio of helium is $k = c_p/c_v = 1.667$.

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Ex. 4. A Carnot heat engine receives heat from a reservoir at 900°C at a rate of 800 kJ/min and rejects the waste heat to the ambient air at 27°C . The entire work output of the heat engine is used to drive a refrigerator that removes heat from the refrigerated space at -5°C and transfers it to the same ambient air at 27°C . Determine (a) the maximum rate of heat removal from the refrigerated space and (b) the total rate of heat rejection to the ambient air.

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Ex. 5. Consider the following three-step process cycle: Heat is allowed to flow out of an ideal monoatomic gas at constant volume so that its pressure drops from 2.5 atm to 1.5 atm. Then the gas expands at constant pressure, from a volume of 6.0 L to 10.0 L, where the temperature reaches its original value. The gas then moves along the isotherm back to its starting point. (a) Draw a P - V diagram showing the three-process cycle. (b) Calculate the total work done by the gas in the process cycle. (c) Calculate the change in internal energy of the gas in the first two processes of the cycle. (d) What is heat flow into or out of the gas on each of the three steps? (e) Is this a heat engine or a refrigerator? What is its efficiency or coefficient of performance? Explain your answer. (f) What is the entropy change for the air in one complete cycle?

[Hint: 1 atm = 1.013×10^5 Pa; 1 L = 1×10^{-3} m³; for a monoatomic gas, $c_v = 3R_u/2$ and $c_p = 5R_u/2$]

Ex. 6. Ambient air at 100 kPa and 300 K is compressed isentropically in a steady-flow device to 0.8 MPa. Determine (a) the work input to the compressor, (b) the exergy of the air at the compressor exit, and (c) the exergy of compressed air after it is cooled to 300 K at 0.8 MPa pressure. Take the gas constant of air to be $R_g = 0.287$ kJ/kg K, the constant pressure specific heat $c_p = 1.005$ kJ/kg K and specific heat ratio $\kappa = 1.4$.

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*Ex. 7. As shown on the T - S diagram below, an ideal gas is adiabatically expanded from State 1 to State 2. T is the temperature and S is the entropy. Show the technical work W_t on this diagram with a detailed explanation. (Hint: The first law of thermodynamics can be written as $Q = \Delta H + W_t$, where Q denotes the heat and ΔH the change of enthalpy.)

